

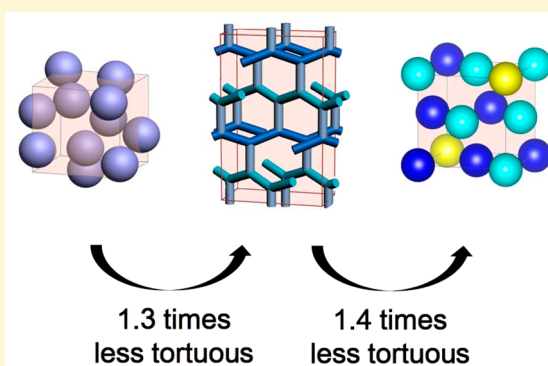
Building Less Tortuous Ion-Conduction Pathways Using Block Copolymer Electrolytes with a Well-Defined Cubic Symmetry

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S Supporting Information

ABSTRACT: We investigated the effects of morphology on the ion transport properties of polymer electrolytes. A single sulfonated block copolymer displayed a series of ordered morphologies with three-dimensional symmetry, i.e., cubic, $Fddd$ (O^{70}), and $Fm\bar{3}m$ (fcc) phases, upon the addition of ionic liquids. The unique phase behavior was understood on the basis of the selective swelling of the sulfonated blocks by ionic liquids and the changes in segregation strength with the modulation of the ionic interactions in the ionic phases. The type of three-dimensional lattice was revealed to play an important role in determining the ion transport properties of ionic liquid-containing sulfonated block copolymers. For example, the sample with disordered spherical lattices exhibited the highest tortuosity (~ 2) for ion conduction, indicative of the considerable diffusion barriers in the conducting phases. On the contrary, the samples with well-defined orthorhombic and face-centered cubic symmetries, i.e., O^{70} and fcc phases, revealed the reductions in tortuosity to 1.52 and 1.17, respectively, attributed to the large grain sizes and large cross-sectional areas of the conducting pathways at grain boundaries. The least tortuous ion conduction seen for fcc phases was particularly remarkable. This unprecedented understanding of the advantage of three-dimensional morphologies with cubic symmetry will open a new avenue toward designing future polymer electrolytes with improved ion transport properties for their uses in diverse applications such as high-temperature fuel cells, batteries, and electro-active actuators.



INTRODUCTION

Polymer electrolytes are emerging as a new class of soft materials that are viable alternatives to liquid electrolytes for a wide range of electrochemical devices, including lithium ion batteries,¹ fuel cells,² and dye-sensitized solar cells.³ Extensive efforts have been devoted to the development of more efficient polymer electrolytes through the tailoring of molecular designs of new polymers with desired functionalities.^{4–7} Nevertheless, the performance of polymer electrolytes is lacking, primarily because of their low conductivity.⁸

The integration of solid salts (e.g., lithium salts)¹ or molten salts (e.g., ionic liquids)^{2,9} into the polymers has thus attracted significant attention because of the improvement of their ion transport properties. In particular, ionic liquids have been considered appealing multifunction ingredients because of their intrinsic natures characterized by high conductivity and high electrochemical and thermal stability.^{10,11} The design of polymers into block architectures composed of ionophilic blocks and mechanically robust ionophobic blocks has also received concomitant attention as a promising strategy for the synergetic achievement of high conductivity and good mechanical integrity from polymer electrolytes comprising ionic liquids.^{1,5,12} This achievement is attributed to the effective

confinement of the ionic liquids within microphase-separated ionic phases.

The most widely investigated block copolymers for this purpose are poly(styrene-*b*-ethylene oxide) (PS-PEO),¹³ poly(styrene-*b*-methyl methacrylate) (PS-PMMA),^{14,15} poly(styrene-*b*-4-vinylpyridine) (PS-P4VP),^{12,16,17} and poly(styrenesulfonate-*b*-methylbutylene) (PSS-PMB).^{18–21} These block copolymers integrated with ionic liquids have been observed to adopt various self-assembled morphologies such as spheres, hexagonally packed cylinders, bicontinuous gyroids, and lamellae by changing the ionic liquid concentration and/or block composition,^{15–22} which can be well located on the trajectory of conventional phase diagrams of nonionic block copolymers.²³

Several studies have reported the ion transport properties of ionic liquid-containing block copolymers, which are affected by factors that include the ionic liquid concentration,^{21,24} the type of cation and anion in the ionic liquid,^{20,25} and the binding affinity of the ionic liquid with the ionophilic polymer.^{20,26} In most cases, the resulting morphologies of block copolymers

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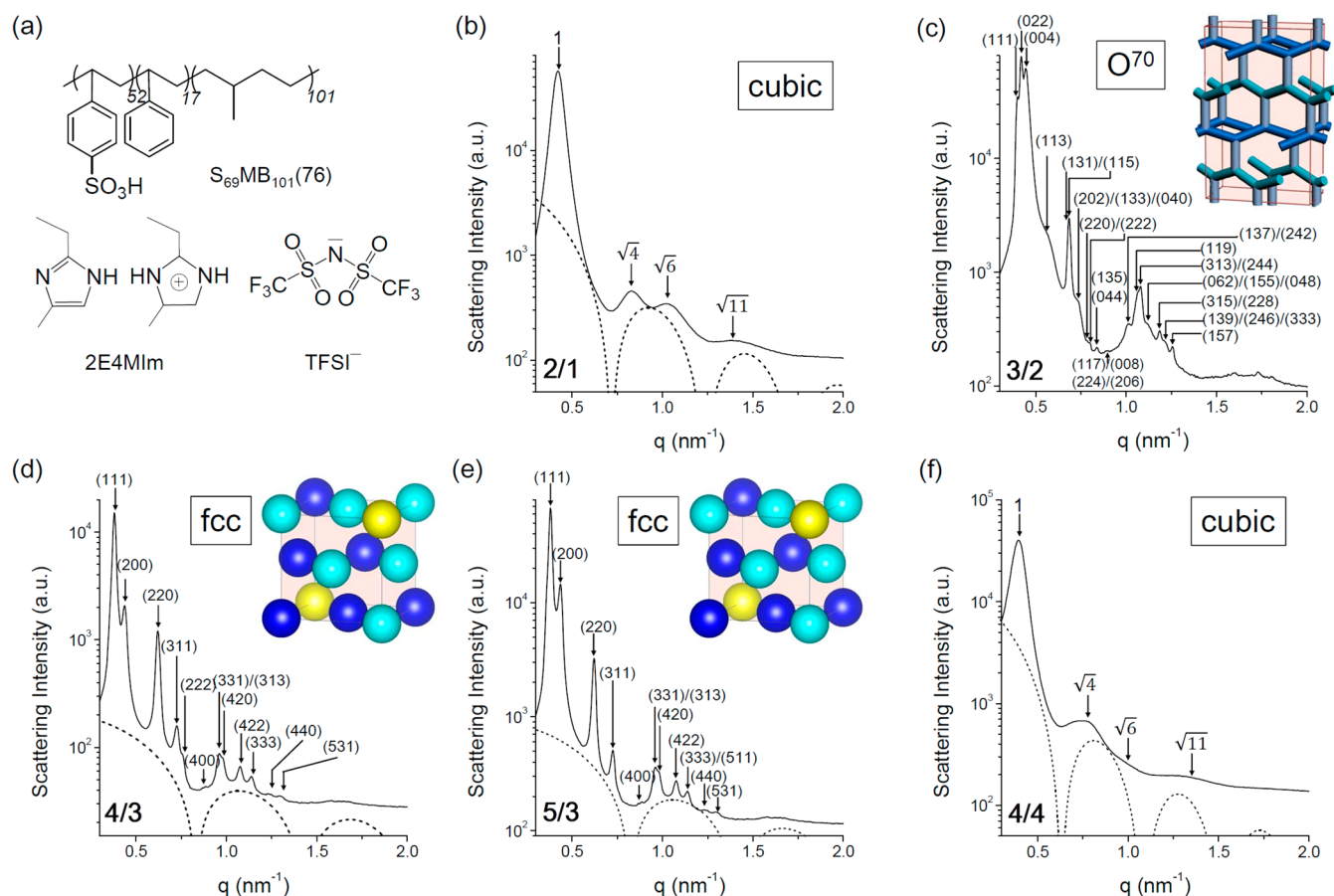


Figure 1. (a) Molecular structures of $S_{69}MB_{101}(76)$, 2E4MIm, and TFSI⁻. Note that the sulfonated PS and nonsulfonated PS are randomly sequenced. (b–f) SAXS profiles of $S_{69}MB_{101}(76)$ copolymers comprising 2E4MIm/TFSI⁻ at ratios of 2/1, 3/2, 4/3, 5/3, and 4/4, respectively, measured at 25 °C. The arrows (↓) represent the series of Bragg peaks of each sample, as indexed in the figure. The dashed lines in panels b and d–f display spherical form factor analysis.

integrated with ionic liquids are lamellar and cylindrical structures as they occupy a large window of the phase diagrams.²⁷ Although such two-dimensional phases are of importance as potential membrane materials, on the basis of a quantitative analysis of the morphology–transport relationship, there is a general consensus that the creation of three-dimensional, less tortuous long-range contiguous paths for ion conduction would be beneficial for improving the ion transport properties.^{28,29}

We previously investigated the phase behavior of a set of PSS-PMB block copolymers upon the addition of various ionic liquids.^{2,18–21} The ionic liquids were observed to selectively solvate the PSS blocks of the PSS-PMB copolymers, causing an increase in the segregation strength of the microphase-separated morphologies. Thus, the PSS-PMB block copolymers comprising ionic liquids underwent a series of phase transitions among disorder, hexagonal cylinder, perforated lamellae, and lamellae.² The fine-tuning of the thermodynamic interactions between the ionic liquids and PSS blocks further allowed us to improve the ion transport efficiency through the implementation of well-defined bicontinuous gyroid morphology.²⁰

Beyond bicontinuous phases, self-assembled morphologies based on spherical lattices with a majority of ionic domains could be of potential interest in the development of continuous ion-conduction pathways.³⁰ In addition, spherical lattices with three-dimensional symmetry offer mechanical strengths higher

than those of two-dimensional lattices.³¹ To this end, a few studies of spherical domains based on ionic liquid-containing block copolymers have been reported;^{17,22} however, the major difficulty in systematic studies is the narrow window of spherical morphologies in phase diagrams. Until recently, the reported spherical morphologies have been limited to body-centered cubic (bcc) phases, and the effects of spherical lattices on the ion transport properties are not well-documented; therefore, a fundamental understanding of this phenomenon is required.

Here we report rich self-assembled morphologies with three-dimensional symmetry, i.e., disordered spherical lattices, an orthorhombic network (O^{70}), and face-centered cubic (fcc) phases, for a single PSS-PMB block copolymer with the addition of ionic liquids. We investigate the ionic conductivity of PSS-PMB block copolymers comprising ionic liquids, with a focus on how the three-dimensional order affects the ion transport efficiency of polymer electrolytes.

EXPERIMENTAL SECTION

Synthesis of a PSS-*b*-PMB Copolymer. A poly(styrene-*b*-methylbutylene) (PS-PMB) block copolymer (7.1–7.1 kg/mol) was synthesized by sequential anionic polymerization of styrene and isoprene and subsequent hydrogenation of the polyisoprene. The molecular weight and molecular weight distribution of the PS-PMB copolymer were characterized by combining ¹H nuclear magnetic resonance (¹H NMR, Bruker AVB-300) spectroscopy with CDCl₃ and

gel permeation chromatography (GPC, Waters Breeze 2 HPLC). The polydispersity index of the PS-PMB block copolymer was 1.03, measured on GPC with polystyrene standards in tetrahydrofuran (THF) for calibration. A flow rate of 1 mL/min was used. The PS block of the PS-PMB copolymer was then sulfonated using procedures described in ref 20 to yield a PSS-PMB copolymer. A sulfonation level (SL) of 76 mol % was determined by ^1H NMR spectra with acetone- d_6 .

Ionic Liquids. 2-Ethyl-4-methylimidazole (2E4MIm, $\geq 95.0\%$), and bis(trifluoromethane) sulfonimide (HTFSI, $\geq 95\%$) were purchased from Sigma-Aldrich and used as received. A set of ionic liquids was synthesized by mixing 2E4MIm and HTFSI in different molar ratios, followed by heating above the melting temperature of the ionic liquid. The final compositions of the synthesized ionic liquids were determined by Fourier transform infrared (FT-IR) spectroscopy experiments using a Spectrum Two IR spectrometer (PerkinElmer), equipped with a He–Ne laser.

Preparation of Ionic Liquid-Containing PSS-PMB Membranes. Inhibitor-free anhydrous tetrahydrofuran (THF, $\geq 99.9\%$) was used without further purification, and methanol was degassed twice before being used. Predetermined amounts of ionic liquids based on 2E4MIm and HTFSI were integrated into the PSS-PMB copolymer at 2E4MIm/HTFSI ratios of 5/3, 4/4, 4/3, 4/2, 3/2, and 2/1 with respect to the moles of $-\text{SO}_3\text{H}$ in the polymer. Namely, the 2E4MIm/HTFSI/ $-\text{SO}_3\text{H}$ mole ratios in the samples were 5/3/1, 4/4/1, 4/3/1, 4/2/1, 3/2/1, and 2/1/1. Then, 5 wt % solutions of the mixtures were prepared using 50/50 vol % THF and methanol mixtures, which were stirred overnight at room temperature. Membranes were prepared by solvent casting under an Ar atmosphere for 2 days followed by vacuum drying at 70°C for 7 days. To exclude the issue of water contamination of the hygroscopic samples, the sample preparations and measurements were performed under an Ar-filled glovebox with a moisture concentration of <0.1 ppm.

Small Angle X-ray Scattering (SAXS). Synchrotron SAXS measurements of the ionic liquid-containing PSS-PMB membranes were performed using the PLS-II 4C SAXS beamline at the Pohang Accelerator Laboratory (PAL). The samples were laminated into an airtight cell to avoid the effects of moisture, and the sample temperature was controlled within $\pm 0.2^\circ\text{C}$ using a sample stage provided by the PLS. The wavelength (λ) of the incident X-ray beam was 0.15 nm ($\Delta\lambda/\lambda = 10^{-4}$), and a sample-to-detector distance of 1.5 m was used. The resulting two-dimensional scattering data were averaged azimuthally to obtain intensity versus scattering wave vector q [$q = 4\pi \sin(\theta/2)/\lambda$, where θ is the scattering angle].

Conductivity Measurements. The conductivities of ionic liquid-containing PSS-PMB membranes were measured by combining a potentiostat (PARSTAT 2273) and AC impedance spectroscopy (Solartron 1260A) in an Ar-filled glovebox. The through-plane conductivity was measured using a home-built two-electrode cell with $1.25\text{ cm} \times 1.25\text{ cm}$ stainless steel blocking electrodes, Kapton spacers, and $1\text{ cm} \times 1\text{ cm}$ Pt working and counter electrodes. The counter electrode was engraved with a $0.8\text{ cm} \times 0.8\text{ cm} \times 200\text{ }\mu\text{m}$ hole to load the samples. A schematic illustration of the conductivity cell is provided in Figure S1 of the Supporting Information. Data were collected over a frequency range of $10\text{--}100000\text{ Hz}$.

Transmission Electron Microscopy (TEM) and Scanning Probe Microscopy (SPM). The 2E4MIm/TFSI-containing PSS-PMB block copolymers were cryo-microtomed at -120°C to obtain thin sections with thicknesses in the range of $80\text{--}120\text{ nm}$ using a RMC Boeckeler PT XL Ultramicrotome. The electron contrast in the samples was enhanced by exposure to ruthenium tetroxide (RuO_4) vapor for 30 min. Imaging of stained samples was performed with a Hitach H-800 microscope operating at 80 kV . Surface morphologies of spin-coated samples on Si wafers were determined using a Nanoscope III (Digital Instrument Inc.).

RESULTS AND DISCUSSION

Figure 1a shows the molecular structure of a PSS-PMB block copolymer, hereafter termed $\text{S}_{69}\text{MB}_{101}(76)$, where the sub-

scripts indicate the degree of polymerization (N) of each block and the number in parentheses indicates the sulfonation level (SL) of the PSS block. Note that the neat $\text{S}_{69}\text{MB}_{101}(76)$ copolymer showed hexagonally packed PMB cylinders dispersed in a PSS matrix (see Figure S2 of the Supporting Information). A set of ionic liquids based on 2-ethyl-4-methylimidazole (2E4MIm) and bis(trifluoromethane)-sulfonimide (TFSI $^-$) was prepared by varying the composition. The chemical structures of the ionic liquids based on 2E4MIm (excessive) and TFSI $^-$ are also shown in Figure 1a. The thermal stability of the 2E4MIm/TFSI $^-$ ionic liquids with various compositions was confirmed by thermogravimetric analysis (TGA) curves (Figure S3 of the Supporting Information).

The synthesized ionic liquids were combined into $\text{S}_{69}\text{MB}_{101}(76)$ at 2E4MIm/TFSI $^-$ mole ratios of 5/3, 4/4, 4/3, 4/2, 3/2, and 2/1 with respect to the moles of the $-\text{SO}_3\text{H}$ group in polymer. The morphologies of the $\text{S}_{69}\text{MB}_{101}(76)$ copolymers containing ionic liquids at different compositions were investigated using small angle X-ray scattering (SAXS) experiments. SAXS profiles of the samples measured at 25°C are presented in Figure 1b–f. No thermotropic phase transitions were observed in the temperature range of $25\text{--}180^\circ\text{C}$ for any of the samples examined, but heating the samples caused gradual increases in domain size of approximately $1\text{--}3\%$ (Figure S4 of the Supporting Information). The composition of the ionic liquid is noted in each figure. At a 2E4MIm/TFSI $^-$ mole ratio of 2/1, we observe Bragg peaks at $1q^*:\sqrt{4}q^*:\sqrt{6}q^*:\sqrt{11}q^*$ with $q^* = 2\pi/d$ and $d = 14.8\text{ nm}$ (Figure 1b), signaling a spherical morphology in the ill-defined cubic lattices. This was confirmed by birefringence measurements on the sample, where the negligible birefringence signals were attained, indicative of a random grain orientation of the isotropic sample (Figure S5 of the Supporting Information). The dashed line in Figure 1b corresponds to a spherical form factor for $R = 6.3\text{ nm}$: $P(q,R) = \{4\pi R^3[\sin(qR) - qR \cos(qR)]/(qR)^3\}^2$ estimated from oscillations of q values with minima near 0.72 and 1.24 nm^{-1} .

The addition of 3/2 2E4MIm/TFSI $^-$ to $\text{S}_{69}\text{MB}_{101}(76)$ (Figure 1c) resulted in the development of more complex Bragg peaks, which can be indexed as the (111), (022), (004), (113), (131), (115), (202), (133), (040), (220), (222), (135), (044), ..., planes of the orthorhombic network symmetry with space group O^{70} (or $Fddd$). The best-fit lattice parameters are $c/a = 3.04$, $a = 18.5\text{ nm}$, and $\theta = 55.7^\circ$. With further changes in the 2E4MIm/TFSI $^-$ mole ratio to 4/2, 4/3, and 5/3, the samples exhibit classical signatures of an fcc phase with the $Fm\bar{3}m$ space group, as demonstrated by the Bragg peaks at $\sqrt{3}q^*:\sqrt{4}q^*:\sqrt{8}q^*:\sqrt{11}q^*:\sqrt{12}q^*:\sqrt{16}q^*:\sqrt{19}q^*:\sqrt{20}q^*:\sqrt{24}q^*:\sqrt{27}q^*$..., with $q^* = 2\pi/d_{111}$ ($d_{111} = 16.6 \pm 1\text{ nm}$). Representative SAXS profiles obtained with 2E4MIm/TFSI $^-$ ratios of 4/3 and 5/3 are shown in panels d and e of Figure 1, respectively. The reflections from (400) planes of the fcc crystals are found to be largely reduced because of the nearby form factor minimum. These results indicate that the window of fcc phases in the phase diagram is quite wide for $\text{S}_{69}\text{MB}_{101}(76)$ copolymers containing 2E4MIm/TFSI $^-$ ionic liquids. It should be noted that the phase transition from O^{70} to fcc phases was accompanied by the substantial increase in d_{111} from 15.8 to 16.6 nm .

Interestingly, the addition of 4/4 2E4MIm/TFSI $^-$ with an equivalent ratio of cation and anion to the $\text{S}_{69}\text{MB}_{101}(76)$ copolymer (Figure 1f) resulted in the re-emergence of disordered spheres, accompanied by a noticeable reduction in

Table 1. Ionic Liquid Loadings, Volume Fractions of PMB Phases, Results of Form Factor Analysis, and Morphologies for S₆₉MB₁₀₁(76) Copolymers Containing Ionic Liquids

sample	ionic liquid loading (wt %)	ϕ_{PMB} (vol %) ^a	ϕ_{PMB} (vol %) ^b	R (nm) ^b	morphology	lattice parameter (nm)
S ₆₉ MB ₁₀₁ (76) with 2/1 2E4MIm/TFSI [−]	60	22	32	6.3	disordered spheres	$a = 14.8^c$
S ₆₉ MB ₁₀₁ (76) with 3/2 2E4MIm/TFSI [−]	73	16	—	—	O ⁷⁰	$a = 18.5$, $c/a = 3.04$, $\theta = 55.7^{ad}$
S ₆₉ MB ₁₀₁ (76) with 4/2 2E4MIm/TFSI [−]	75	14	13	5.1	fcc	$a = 25.6^e$
S ₆₉ MB ₁₀₁ (76) with 4/3 2E4MIm/TFSI [−]	79	13	11	5.4	fcc	$a = 28.6^e$
S ₆₉ MB ₁₀₁ (76) with 5/3 2E4MIm/TFSI [−]	81	11	10	5.3	fcc	$a = 28.8^e$
S ₆₉ MB ₁₀₁ (76) with 4/4 2E4MIm/TFSI [−]	82	11	38	6.8	disordered spheres	$a = 15.9^e$

^aEstimated from densities of pure components. ^bEstimated from form factor analysis. ^c $q_{hkl} = 2\pi(h^2 + k^2 + l^2)^{1/2}/a$, assuming that the first peak of the SAXS profile is q_{100} . h , k , and l are Miller indices. ^d $q_{hkl} = 2\pi[(h/a)^2 + (k/b)^2 + (l/c)^2]^{1/2}$. ^e $q_{hkl} = 2\pi(h^2 + k^2 + l^2)^{1/2}/a$. The first peak of the SAXS profile is q_{111} .

domain spacing to 15.9 nm. This finding indicated that the lyotropic phase transitions now passed order–disorder phase boundaries. From spherical form factor analysis, the radius of the PMB sphere was determined to be increased by >20%, compared to that of the fcc-forming samples. The results of form factor analysis are summarized in Table 1.

To calculate the volume fraction of PMB domains in the ionic liquid-containing S₆₉MB₁₀₁(76) copolymers, ϕ_{PMB} , we employed two approaches. First, by using pure component densities of $\rho_{\text{PS}} = 1.05 \text{ g/cm}^3$, $\rho_{\text{PSS}} = 1.44 \text{ g/cm}^3$, $\rho_{\text{PMB}} = 0.86 \text{ g/cm}^3$, $\rho_{2\text{E4MIm}} = 0.98 \text{ g/cm}^3$, and $\rho_{\text{TFSI}} = 1.94 \text{ g/cm}^3$, the estimated ϕ_{PMB} values were in the range of 11–22 vol %, ignoring volume changes of mixing. Second, the ϕ_{PMB} values were calculated from the form factor analysis using equation $\phi_{\text{PMB}} a^3 = N \times 4/3 \pi R^3$, assuming that all the chains are involved in the self-assembly, where N is the number of spheres in a unit cell, R is the radius of the PMB sphere, and a is the lattice parameter of the unit cell measured by SAXS experiments.³² This yields ϕ_{PMB} values in the range between 10 and 38 vol %, as summarized in Table 1. For fcc phases, the ϕ_{PMB} values estimated from the form factor analysis were similar to those based on density calculation.

If we use the qualitative assumption that changing the 2E4MIm/TFSI[−] composition from 2/1 to 3/2, 4/2, 4/3, 5/3, and 4/4 corresponds to a simple increase in the ionic liquid concentration in S₆₉MB₁₀₁(76)/ionic liquid composite membranes, the PSS blocks are expected to be increasingly swollen and the interfacial curvature is gradually driven toward the ionophobic PMB domains. In this case, the projected phases follow a horizontal path across the conventional phase diagrams of nonionic block copolymers toward increasing block volume fraction,²³ which is, in part, consistent with our observation. However, because spherical lattices other than bcc phases are known to occupy a very narrow window of the phase diagram, it is inferred that the addition of ionic liquids to an ionic block copolymer greatly expanded the range of accessible self-assembled morphology by modifying the balance of interfacial area and chain stretching.

This finding is analogous to the results of nonionic block copolymer/selective solvent systems, as pioneered by Lodge et al.³³ One major difference, however, is that the modification of ionic interactions by varying the ratio of cations and anions in ionic liquids is a cause of altering the selectivity of ionic liquids to the PSS phases, which is intimately related to the changes in the segregation strength between PMB domains and ionic liquid-embedded PSS phases. For example, a 4/2 2E4MIm/TFSI[−] system is more selective for the PSS block than a 3/2 2E4MIm/TFSI[−] system, such that the chain stretching to reduce the interfacial area is sufficiently great to induce a phase

transition. The decrease in segregation strength in ionic liquid-containing PSS-PMB block copolymers is inferred when the composition is closer to the equivalence, as seen from the smallest swelling of PSS phases ($1 - \phi_{\text{PMB}}$) with a 4/4 2E4MIm/TFSI[−] system. The segregation strength of the microphase-separated morphology can also be modulated by the variation of the type of ionic liquid, which has not been examined in this study. More quantitative analysis of the thermodynamic parameters of ionic liquid-containing PSS-PMB block copolymers will be conducted in future work. Note that all ionic liquids employed in this study are selective for PSS phases, as confirmed by monitoring the glass transition temperatures of PSS and PMB chains upon the addition of ionic liquids (data not shown here).

In the literature, the most widely studied block copolymers demonstrating rich three-dimensional phases are nonionic linear triblock copolymers, i.e., polystyrene-*b*-polybutadiene-*b*-poly(methylmethacrylate) (SBM) and polyisoprene-*b*-polystyrene-*b*-poly(ethylene oxide) (ISO), as pioneered by Stadler et al.³⁴ and Bates et al.,³⁵ respectively. For example, the phase diagrams of ISO systems enclose O⁷⁰ phases, which are bordered by cubic symmetries of the core–shell gyroid (O²³⁰) and alternating gyroid (Q²¹⁴). In this study, the gyroid microstructure is not definitively observed for any S₆₉MB₁₀₁(76) copolymers integrated with a 2E4MIm/TFSI[−] system in a range of ionic liquid compositions. However, it should be noted that our previous studies of the S₃₀MB₄₄(17) copolymer with a low molecular weight and a low SL, i.e., weaker segregation (low χN , where χ is the Flory–Huggins interaction parameter and N the degree of polymerization), revealed well-defined Q²³⁰ phases (*Ia* $\bar{3}d$ space group) upon the addition of a 2/1 2E4MIm/TFSI[−] system.²⁰ This implies that the high χN – small ϕ_{PMB} window of the phase diagram is taken up by the O⁷⁰ and fcc phases, whereas Q²³⁰ phases become stable with a decrease in χN for our sulfonated block copolymers containing ionic liquids.

The morphology–transport relationship of the ISO triblock copolymers (with the inclusion of cross-linkable moieties) with the addition of ionic liquids has also been investigated by Lodge and co-workers.³⁶ However, the three-dimensional morphologies were readily transformed into two-dimensional hexagonal cylindrical structures even with a small amount of ionic liquid. Consequently, to date, no study has reported the effects of three-dimensional symmetry on the ion transport properties of block copolymer electrolytes. In contrast, the rich phase behavior of S₆₉MB₁₀₁(76) copolymers comprising ionic liquids allowed us to investigate the advantages or disadvantages of three-dimensional symmetry for ion transport that have

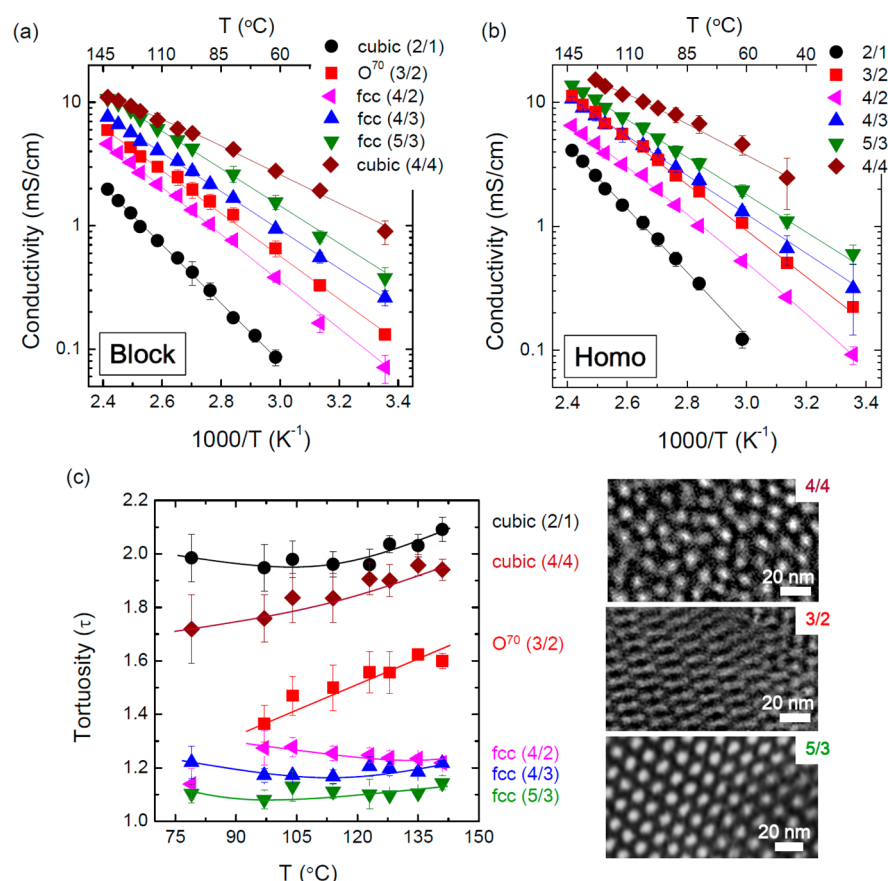


Figure 2. Through-plane conductivities of ionic liquid-containing (a) $S_{69}MB_{101}(76)$ and (b) $S_{74}(75)$ at various compositions, as indicated in the figure. The ionic liquid compositions listed in panels a and b are in order of increasing ionic liquid content. Solid lines indicate analysis using the VTF equation. (c) Dissimilar tortuosities depending on the type of three-dimensional phases predicted using eq 1. Solid lines through the data represent the best regression curves. Representative TEM images obtained with 4/4, 3/2, and 5/3 2E4MIm/TFSI⁻ systems are shown at the right to show the disordered spheres, O^{70} , and fcc phases, respectively. Ionic liquid-containing PSS phases are shown dark by RuO₄ staining.

ultimately allowed us to unveil the ideal morphology with less tortuous ion-conduction pathways.

Figure 2a shows the ionic conductivity of $S_{69}MB_{101}(76)$ combined with 2E4MIm/TFSI⁻ systems for various compositions in the temperature range of 25–150 °C. As controls, the results of ionic liquid-containing PSS homopolymers are also shown in Figure 2b wherein a PSS homopolymer with $N = 74$ and SL = 75 mol % [hereafter $S_{74}(75)$] was employed. Analogous to the $S_{69}MB_{101}(76)$ block copolymer, we preserved the local concentration of ionic liquids within the PSS phases [the 2E4MIm/HTFSI-/SO₃H mole ratios in ionic liquid-containing $S_{74}(75)$ homopolymers were 5/3/1, 4/4/1, 4/3/1, 4/2/1, 3/2/1, and 2/1/1]. Therefore, there have been increases in the actual weight fraction of ionic liquids in the homopolymer samples, corresponding to 70–88 wt % (the values are listed in Table S1 of the Supporting Information).

The ionic conductivity of composite membranes appears to be closely related to the intrinsic transport properties of neat ionic liquids. In other words, a high ionic conductivity was achieved with a 4/4 2E4MIm/TFSI⁻ system for both $S_{69}MB_{101}(76)$ and $S_{74}(75)$, whereas the conductivity was low for the samples containing a 4/2 2E4MIm/TFSI⁻ system, in qualitative agreement with the conductivity of neat ionic liquids, as summarized in Table 2. However, the opposite trend in conductivity between composite membranes and neat ionic liquids was seen with the 5/3 and 4/3 2E4MIm/TFSI⁻ compositions. This indicates that the ion transport properties

Table 2. Transport Properties of Neat Ionic Liquids Used in This Study

2E4MIm/TFSI ^{-a} composition	ionic conductivity (mS/cm) ^b		activation E (kJ/mol) ^c
	25 °C	100 °C	
4/4	3.0	14.8	18.2 ± 1.0
5/3	1.1	5.0	16.8 ± 1.2
4/3	1.7	10.1	17.9 ± 1.1
3/2	0.89	3.4	15.2 ± 0.8
2/1 (4/2)	0.48	2.2	18.0 ± 0.9

^aThe compositions are in order of decreasing ionic liquid content.

^bMeasured using impedance spectroscopy. ^cObtained with the Arrhenius equation.

of ionic liquids can be significantly altered if they are embedded in sulfonated polymers because of the existence of ionic interactions between polymer matrices and combined ionic liquids. A large reduction in conductivity for the 2/1 2E4MIm/TFSI⁻ samples is attributed to the low ion concentration in the membranes.

We note in passing that the nonstoichiometric 2E4MIm/TFSI⁻ is a eutectic mixture of 2E4MIm and 2E4MIm/TFSI⁻, attributed to the π - π interactions between 2E4MIm molecules in forming the 2E4MIm-HTFSI complexes. Therefore, an arithmetical expectation of the ionic conductivities of neat ionic liquids based on composition is not easy, as can be seen from

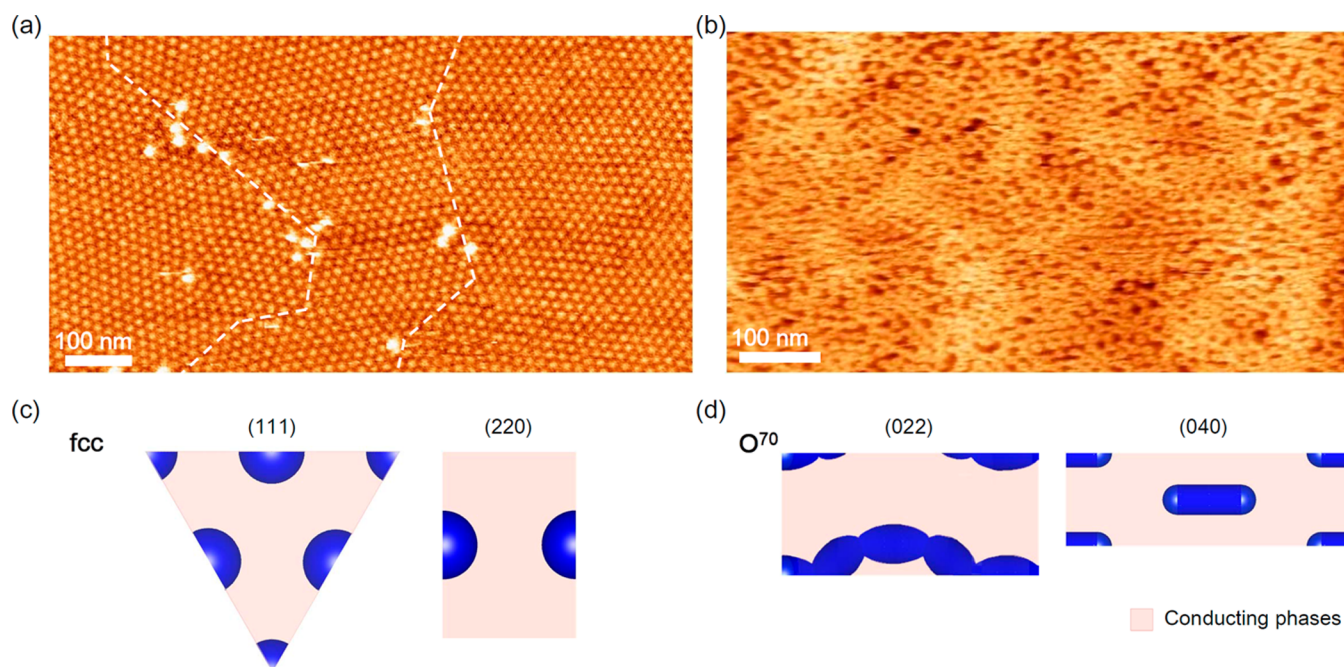


Figure 3. SPM phase images of (a) fcc and (b) O^{70} phases. Dotted lines in panel a represent grain boundaries in fcc phases. Visualizations of the cross-sectional area of (c) fcc and (d) O^{70} unit cells at grain boundaries along different planes.

Table 2. This is consistent with the results of Watanabe et al.³⁷ It should be also noted here that all $S_{69}MB_{101}(76)$ samples combined with $2E4MIm/TFSI^-$ were in hard gel states, contrary to the clear liquid states of $S_{74}(75)$ at the same $2E4MIm/TFSI^-$ composition, ascribed to the improved mechanical properties of the PSS-PMB block copolymer by covalently attaching ionophobic PMB blocks to PSS chains.

Solid lines in panels a and b of Figure 2 represent the analysis using the Vogel–Tamman–Fulcher (VTF) equation, yielding dissimilar potential barriers to ion conduction of 294, 779, 671, 863, 711, and 954 K [for $S_{69}MB_{101}(76)$] and 304, 1197, 1228, 1281, 1205, and 1473 K [for $S_{74}(75)$] for $2E4MIm/TFSI^-$ compositions of 4/4, 5/3, 4/3, 4/2, 3/2, and 2/1, respectively. Because the $\log(\text{conductivity})$ values are very linear with the inverse of temperature, the data were also fit with an Arrhenius equation (fits are not shown here), leading to activation energies (E_a) of 20.3, 28.0, 30.2, 35.7, 32.0, and 46.4 kJ/mol [for $S_{69}MB_{101}(76)$] and 18.7, 28.6, 30.5, 34.7, 33.2, and 49.2 kJ/mol [for $S_{74}(75)$] for $2E4MIm/TFSI^-$ compositions of 4/4, 5/3, 4/3, 4/2, 3/2, and 2/1, respectively. The fact that the tendency of E_a with ionic liquid composition for ionic liquid-containing polymers is markedly different from that of neat ionic liquids (Table 2) is noteworthy.

The conductivity of nanostructured electrolytes (σ) is known to be influenced by the structural parameters, i.e., interfacial area, grain boundaries, and the conducting network topology, which can be quantified by the tortuosity (τ) as shown in eq 1.³⁸

$$\sigma = \frac{\phi \sigma_{\max}}{\tau} \quad (1)$$

where ϕ and $\sigma_{\max}$ denote the volume fraction and ionic conductivity of the conducting phases, respectively.

To underpin the effects of the three-dimensional symmetry on the ion transport properties of $S_{69}MB_{101}(76)$ copolymers combined with $2E4MIm/TFSI^-$, the τ values were predicted

using eq 1 and are shown in Figure 2c. The type of morphology and composition of $2E4MIm/TFSI^-$ are noted in the figure. The ϕ_{PMB} values estimated from densities of pure components (Table 1) were used to calculate $\phi (=1 - \phi_{PMB})$, and the ionic conductivity of ionic liquid-containing $S_{74}(75)$ samples (Figure 2b) was used as $\sigma_{\max}$. The samples containing 2/1 and 4/4 $2E4MIm/TFSI^-$ systems exhibited the highest tortuosity (1.93 ± 0.17), indicative of considerable dead ends in the conducting phases and/or diffusion barriers at grain boundaries for the disordered spherical lattices, which are commonly cited problems for self-assembled polymer electrolytes. The τ value of O^{70} phases at a 3/2 $2E4MIm/TFSI^-$ ratio is approximately 1.52 ± 0.12 in the temperature window of interest. The reduction in tortuosity observed for O^{70} phases, compared with disordered spheres, can be rationalized by co-continuous PSS domains, allowing efficient ion conduction across the membrane. The noticeable temperature-dependent tortuosity of O^{70} phases unlike that of other samples may be ascribed to grain melting and/or interfacial mixing with heating.

Most strikingly, the samples comprising 4/2, 4/3, and 5/3 $2E4MIm/TFSI^-$ showed the lowest τ values of 1.17 ± 0.08 wherein the corresponding morphology was fcc phases. This leads us to conclude that well-defined spherical morphologies based on face-centered cubic symmetry with majority ionic phases may be important to the improvement of ion transport properties of polymer electrolyte membranes. Transmission electron microscopy (TEM) images shown in Figure 2c confirm the disordered spheres, O^{70} , and fcc phases of the samples.

Given that the tortuosity quantified the topological parameters of the conducting phases, possible explanations for the different tortuosities for the O^{70} and fcc phases despite the similar volume fractions of the conducting phases (84 and 87%, respectively) include the different grain sizes and dissimilar cross-sectional areas of the conducting pathways at grain boundaries. We studied $S_{69}MB_{101}(76)$ samples combined with 5/3 and 3/2 $2E4MIm/TFSI^-$ systems wherein the

corresponding morphologies were fcc and O^{70} phases, respectively. Scanning probe microscopy (SPM) equipped with an environmental chamber was employed for grain size analysis by preparing 250 nm thick thin films on Si wafers. This is to exclude the issue of water contamination of hygroscopic samples during sample preparation, i.e., staining for TEM experiments.

The surface morphologies of annealed films, examined in SPM tapping mode, are shown in panels a and b of Figure 3. A grain orientation having 3-fold [111] symmetry with relatively small defect densities was evident for the fcc phases. On the other hand, the O^{70} phases were found out to be defective, which should be responsible for the increased τ values rather than the theoretical value of 1. In panels c and d of Figure 3, we further present visualizations of the cross-sectional areas of fcc and O^{70} unit cells, respectively, at representative grain boundaries. On the basis of the lattice parameter of each unit cell measured by SAXS experiments (Figure 1 and Table 1), it has been revealed that the conducting phases occupy 76 and 85% of the cross-sectional area for the fcc phases at planes of (111) and (220), respectively. In contrast, the areas were as small as 70 and 74% for the O^{70} phases at the (022) and (040) planes, respectively. Consequently, the higher-efficiency ion conduction across the fcc phases can be rationalized by the large grain sizes and large cross-sectional areas of the conducting pathways. It is worth noting that the cross-sectional areas become largely reduced to 44 and 61% for simple cubic phases at the (100) and (110) planes, respectively. This should be closely associated with the highest tortuosity for the samples with 2E4MIm/TFSI⁻ ratios of 4/4 and 2/1, which showed disordered spherical lattices.

On the basis of the results obtained thus far, we present the schematic illustration of dissimilar ion transport efficiencies of ionic liquid-containing PSS-PMB copolymers, depending on the three-dimensional order, in Figure 4. With a majority of

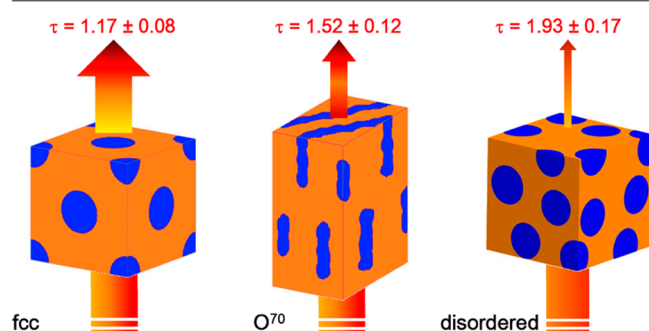


Figure 4. Schematic illustration of the effects of three-dimensional morphology on the ion transport properties of ionic liquid-containing sulfonated block copolymers.

ionic PSS phases, fast ion conduction can be achieved when the sample displays well-defined cubic symmetries, whereas a reduction in ion transport efficiency was demonstrated for spherical lattices lacking organization. The fcc phases with cubic symmetry appeared to be preferable over O^{70} phases for achieving nontortuous ion-conduction pathways. It should be noted that previously reported tortuosity values lie in range between 1.5 and 3, not only for the conductivity of block copolymer electrolytes but also for diffusion of small molecules in disordered, co-continuous network morphologies.^{38–40} This suggests unprecedented advantages of ionic liquid-containing

PSS-PMB copolymers having well-defined cubic symmetry to improve ion transport properties.

CONCLUSION

We have presented the first experimental data unveiling the ideal three-dimensional morphology of block copolymer electrolytes for creating less tortuous ion-conduction pathways. A range of three-dimensional phases were developed for a single PSS-PMB block copolymer by incorporating a range of nonstoichiometric ionic liquids. For example, the addition of a 3/2 2E4MIm/TFSI⁻ system into the PSS-PMB copolymer resulted in the development of O^{70} phases with orthorhombic symmetry, whereas with 4/2, 4/3, and 5/3 2E4MIm/TFSI⁻ compositions, well-defined fcc phases with cubic symmetry were formed. To some extent, the framework of conventional phase diagrams of block copolymers/selective solvents could be applied to our observation. The structural optimization of ionic liquid-containing membranes was crucial for attaining desired ion transport properties; the direct comparison of disordered spheres, O^{70} , and fcc phases revealed the efficient ion transport in face-centered cubic phases over orthorhombic network phases. This intriguing observation was rationalized by the dissimilar topological parameters of the conducting phases, which depended on the three-dimensional symmetry.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.chemmater.5b04157.

Characterization of the PSS-PMB block copolymer and the ionic liquids (PDF)

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Notes

The authors declare no competing financial interest.

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